

Software Requirements Specification (SRS) for the **Love Maths** example project

A learning aid for the software testing unit

1	The game is aimed towards children and adults who are looking to improve their proficiency with mental arithmetic. The player should be reasonably Internet literate, and be able to use a web browser, mouse and keyboard. The game expects players to already know how to perform the operations and will not seek to explain the maths. Instead, the focus is only on providing them with questions they could practise, and let them track how well they are doing.
2	The main functions the system must perform are: 1. To provide an interface containing four buttons for the player to choose which arithmetic operation they want to play with now: addition, subtraction, multiplication and division. The default when the game first loads should be addition. 2. For the chosen operation, the game should offer the player a mathematical challenge with randomised operands for the given operation. 3. The player would then input their answer to the problem and click "Submit Answer" or press the enter key to submit it. 4. Only whole numbers should be accepted or provided. 5. After the player submits an answer, the game should tell the player whether they were correct or not. If they were not correct, it should tell them what the correct answer was.

	- The game should use counters to track the number of correct and incorrect answers for the current game session. After every question, one of these counters should be incremented by 1. - The game should then continuously generate new mathematical challenges using the same mathematical operation.
3	Love Maths is intended to be an online single-player game for practising the four arithmetic operations on natural numbers. This game will allow a player to generate an endless sequence of arithmetic problems (e.g. $2+2$) which the player would need to solve.
4	The non-functional requirements are: - Security - there should be no way for a player to change their score, other than by playing the game. - Performance - every interface operation should take less than 1 second. - Usability - the game's user interface should look good on all common screen sizes.
5	We expect there to be two kinds of players: - Casual players - those who might only play the game occasionally (or even just once). These players would just try it for fun to assess their current proficiency. - Serious players - these players are using the system with a specific goal of improving their proficiency. They might then play the game every day (possibly even have an actual training plan), and track how their proficiency (measured by the game score) improves over time.